Inverse Trigonometric Functions

ISC Class 12 Previous Years Board Questions

with Answers

2000 to Sem 1-2021

1. Prove that : $\sin^{-1}\frac{x}{\sqrt{1+x^2}} + \cos^{-1}\frac{x+1}{\sqrt{x^2+2x+2}} = \tan^{-1}(x^2+x+1)$ (ISC 2000)

2. Solve : $\tan^{-1}(x+2) + \tan^{-1}(2-x) = \tan^{-1}\left(\frac{2}{3}\right)$ (ISC 2001)

3. Show that : $\sin^{-1}\frac{\sqrt{3}}{2} + 2\tan^{-1}\frac{1}{\sqrt{3}} = \frac{2\pi}{3}$ (ISC 2002)

4. Show that $\sin^{-1}\left(\frac{4}{5}\right) + \cos^{-1}\left(\frac{2}{\sqrt{5}}\right) = \cot^{-1}\left(\frac{2}{11}\right)$ (ISC 2003)

5. Prove that $2\left(\tan^{-1}1 + \tan^{-1}\frac{1}{2} + \tan^{-1}\frac{1}{3}\right) = \pi$ (ISC 2004)

6. Prove that $\cot\left(\frac{\pi}{4} - 2\cot^{-1}3\right) = 7$ (ISC 2005)

7. Show that $\sin^{-1}\left(\frac{1}{\sqrt{17}}\right) + \cos^{-1}\left(\frac{9}{\sqrt{85}}\right) = \tan^{-1}\left(\frac{1}{2}\right)$ (ISC 2006)

8. Prove that $2\tan^{-1}\left(\frac{1}{3}\right) + \cot^{-1}(4) = \tan^{-1}\left(\frac{16}{13}\right)$ (ISC 2007)

9. Solve : $\tan^{-1}(x-1) + \tan^{-1}x + \tan^{-1}(x+1) = \tan^{-1}3x$ (ISC 2008)

10. If $\sin^{-1}x + \sin^{-1}y + \sin^{-1}z = \pi$, *prove that* $x^2 - y^2 - z^2 + 2yz\sqrt{1-x^2} = 0$

(ISC 2009)

11.Solve : $\cos^{-1}[\sin(\cos^{-1} x)] = \frac{\pi}{3}$ (ISC 2010)

12. Prove that $\sin\left[2\tan^{-1}\left(\frac{3}{5}\right) - \sin^{-1}\left(\frac{7}{25}\right)\right] = \frac{304}{425}$ (ISC 2010)

13.Solve for x : $\sin(2\tan^{-1} x) = 1$ (ISC 2011)

14.Prove that $2\tan^{-1}\frac{1}{5} + \cos^{-1}\frac{7}{5\sqrt{2}} + 2\tan^{-1}\frac{1}{8} = \frac{\pi}{4}$ (ISC 2011)

15.Prove that $sec^2(\tan^{-1} 2) + cosec^2(\cot^{-1} 3) = 15$ (ISC 2012)

16.Prove that $\cos^{-1}\left(\frac{63}{65}\right) + 2\tan^{-1}\left(\frac{1}{5}\right) = \sin^{-1}\left(\frac{3}{5}\right)$ (ISC 2012)

17. Prove that : $\tan^{-1}\left(\frac{1}{4}\right) + \tan^{-1}\left(\frac{2}{9}\right) = \frac{1}{2}\sin^{-1}\left(\frac{4}{5}\right)$ (ISC 2013)

18.Solve for x : $\sin^{-1} x + \sin^{-1}(1-x) = \cos^{-1} x$ (ISC 2013, 2017)

19.Evaluate : $\tan\left[2\tan^{-1}\frac{1}{2} - \cot^{-1} 3\right]$ (ISC 2014)

20.If $\cos^{-1} x + \cos^{-1} y + \cos^{-1} z = \pi, prove\ that\ \ x^2 + y^2 + z^2 + 2xyz = 1$ (ISC 2014)

21.Solve : $\cos^{-1}(\sin\cos^{-1} x) = \frac{\pi}{6}$ (ISC 2015)

22.Solve for x : $\sin^{-1}\frac{5}{x} + \sin^{-1}\frac{12}{x} = \frac{\pi}{2}\ \ , x \neq 0$ (ISC 2015)

23.Solve for x, if : $\tan(\cos^{-1} x) = \frac{2}{\sqrt{5}}$ (ISC 2016)

24.If $\sin^{-1} x + \tan^{-1} x = \frac{\pi}{2}, prove\ that\ \ 2x^2 + 1 = \sqrt{5}$ (ISC 2016)

25.Prove that $\frac{1}{2}\cos^{-1}\left(\frac{1-x}{1+x}\right) = \tan^{-1}\sqrt{x}$ (ISC 2017)

26.Solve for x : $\sin^{-1} x + \sin^{-1}(1-x) = \cos^{-1} x\,, x \neq 0$ (ISC 2017)

27.Solve for x : $3\tan^{-1}x + \cot^{-1}x = \pi$ (ISC 2018)

28.If $\tan^{-1}a + \tan^{-1}b + \tan^{-1}c = \pi, prove\ that\ a + b + c = abc$ (ISC 2018)

29.Solve for x : $\tan^{-1}\left(\frac{x-1}{x-2}\right) + \tan^{-1}\left(\frac{x+1}{x+2}\right) = \frac{\pi}{4}$ (ISC 2019)

30.If $\sec^{-1}x = \operatorname{cosec}^{-1}y, show\ that\ \frac{1}{x^2} + \frac{1}{y^2} = 1$ (ISC 2019)

31.Prove that : $tan^2(\sec^{-1}2) + cot^2(\operatorname{cosec}^{-1}3) = 11$ (ISC 2020)

32.If $\cos^{-1}\frac{x}{2} + \cos^{-1}\frac{y}{3} = \theta$, then prove that (ISC 2020)

$$9x^2 - 12xy\ \cos\theta + 4y^2 = 36\ sin^2\theta$$

33.Evaluate : $\cos(2\ \cos^{-1}x + \sin^{-1}x)$ at $x = \frac{1}{5}$. (ISC 2020)

34. Choose and select the correct option: (ISC Sem 1-2021)

If $\cot^{-1}\frac{1}{5} + \tan^{-1}x = \frac{\pi}{2}$, then the value of x is

(a) $\frac{1}{5}$

(b) 1

(c) 0

(d) $\frac{-1}{5}$

Answer: _____

35. Choose and select the correct option: (ISC Sem 1-2021)

The expression for $\cos(\tan^{-1}x)$ is equal to:

(a) $\frac{1}{\sqrt{1-x^2}}$

(b) $\frac{1}{\sqrt{1+x^2}}$

(c) $\frac{\sqrt{1-x^2}}{2}$

(d) $\sqrt{1-x^2}$

Answer: _____

36. Choose and select the correct option: (ISC Sem 1-2021)

The value of $\tan^{-1} 1 + \cos^{-1} \frac{1}{2}$ is

(a) $\frac{5\pi}{12}$

(b) $\frac{\pi}{4}$

(c) $\frac{\pi}{2}$

(d) $\frac{7\pi}{12}$

Answer: _____

Answer

[2]	± 3	[9]	$0, \pm\frac{1}{2}$	[11]	$\pm\frac{\sqrt{3}}{2}$	[13]	1
[18]	$0, \frac{1}{2}$	[19]	$\frac{9}{13}$	[21]	$\pm\frac{1}{2}$	[22]	13
[23]	$\frac{\sqrt{5}}{3}$	[27]	1	[29]	$\pm\frac{1}{\sqrt{2}}$	[33]	$-\frac{2\sqrt{6}}{5}$
[34]	(a) $\frac{1}{5}$	[35]	(b) $\frac{1}{\sqrt{1+x^2}}$	[36]	(d) $\frac{7\pi}{12}$		

Inverse Trigonometric Functions

Solution of Previous Years ISC Class 12 Board Questions

2000 to 2020

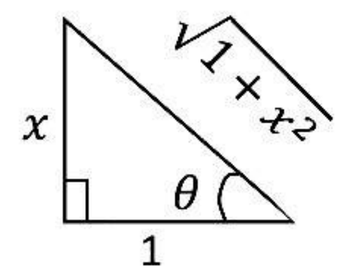
1. Prove that $\sin^{-1}\frac{x}{\sqrt{x^2+1}} + \cos^{-1}\frac{x+1}{\sqrt{x^2+2x+2}} = \tan^{-1}(x^2 + x + 1)$ (ISC 2000)

Solution : Let $\sin^{-1}\frac{x}{\sqrt{1+x^2}} = \theta$

$\Rightarrow \sin\theta = \frac{x}{\sqrt{1+x^2}}$

$\therefore \tan\theta = x$

$\therefore \theta = \tan^{-1} x$

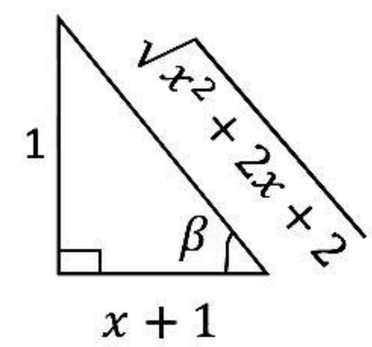


Let $\cos^{-1}\frac{x+1}{\sqrt{x^2+2x+2}} = \beta$

$\therefore \cos\beta = \frac{x+1}{\sqrt{x^2+2x+2}}$

$\therefore \tan\beta = \frac{1}{x+1}$

$\therefore \beta = \tan^{-1}\frac{1}{x+1}$



Now **L.H.S.**

$= \tan^{-1} x + \tan^{-1}\frac{1}{x+1}$

$= \tan^{-1}\left(\frac{x+\frac{1}{x+1}}{1-x.\frac{1}{x+1}}\right)$

$= \tan^{-1}\left(\frac{\frac{x^2+x+1}{x+1}}{\frac{x+1-x}{x+1}}\right)$

$= \tan^{-1}\left(\frac{x^2+x+1}{1}\right)$

$= \tan^{-1}(x^2 + x + 1)$ Proved

2. Solve : $\tan^{-1}(x+2) + \tan^{-1}(2-x) = \tan^{-1}\left(\frac{2}{3}\right)$ (ISC 2001)

Solution : $\tan^{-1}(x+2) + \tan^{-1}(2-x) = \tan^{-1}\left(\frac{2}{3}\right)$

$\Rightarrow \tan^{-1}\left\{\frac{x+2+2-x}{1-(x+2)(2-x)}\right\} = \tan^{-1}\left(\frac{2}{3}\right)$

$\Rightarrow \frac{4}{1-(4-x^2)} = \frac{2}{3}$

$\Rightarrow \frac{2}{x^2-3} = \frac{1}{3}$

$\Rightarrow x^2 - 3 = 6$

$\Rightarrow x^2 = 9$

$\Rightarrow x = \pm 3$ Answer.

3. Show that : $\sin^{-1}\frac{\sqrt{3}}{2} + 2\tan^{-1}\frac{1}{\sqrt{3}} = \frac{2\pi}{3}$ (ISC 2002)

Solution : **L.H.S.** $= \sin^{-1}\frac{\sqrt{3}}{2} + \tan^{-1}\left\{\frac{2.\frac{1}{\sqrt{3}}}{1-\left(\frac{1}{\sqrt{3}}\right)^2}\right\}$

$$= \sin^{-1}\frac{\sqrt{3}}{2} + \tan^{-1}\left(\frac{\frac{2}{\sqrt{3}}}{\frac{2}{3}}\right)$$

$$= \sin^{-1}\frac{\sqrt{3}}{2} + \tan^{-1}\left(\frac{2}{\sqrt{3}} \times \frac{3}{2}\right)$$

$$= \sin^{-1}\frac{\sqrt{3}}{2} + \tan^{-1}\sqrt{3}$$

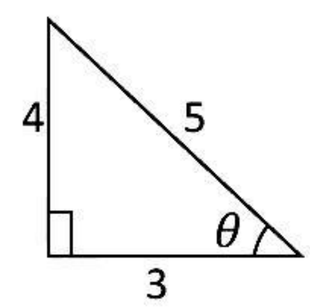
$$= \frac{\pi}{3} + \frac{\pi}{3}$$

$$= \frac{2\pi}{3}$$

Proved

4. Show that $\sin^{-1}\left(\frac{4}{5}\right) + \cos^{-1}\left(\frac{2}{\sqrt{5}}\right) = \cot^{-1}\left(\frac{2}{11}\right)$ (ISC 2002)

Solution : Let $\sin^{-1}\left(\frac{4}{5}\right) = \theta$

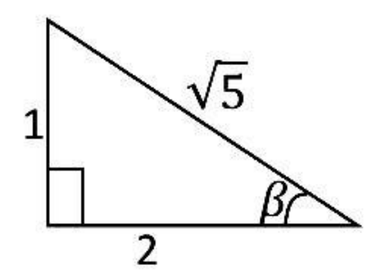


$\therefore \quad \sin\theta = \frac{4}{5}$

$\therefore \quad \tan\theta = \frac{4}{3}$

$\therefore \quad \theta = \tan^{-1}\left(\frac{4}{3}\right)$

Let $\cos^{-1}\left(\frac{2}{\sqrt{5}}\right) = \beta$



$\therefore \quad \cos\beta = \frac{2}{\sqrt{5}}$

$\therefore \quad \tan\beta = \frac{1}{2}$

$\therefore \quad \beta = \tan^{-1}\left(\frac{1}{2}\right)$

L.H.S. $= \sin^{-1}\left(\frac{4}{5}\right) + \cos^{-1}\left(\frac{2}{\sqrt{5}}\right)$

$$= \tan^{-1}\left(\frac{4}{3}\right) + \tan^{-1}\left(\frac{1}{2}\right)$$

$$= \tan^{-1}\left(\frac{\frac{4}{3}+\frac{1}{2}}{1-\frac{4}{3}\times\frac{1}{2}}\right)$$

$$= \tan^{-1}\left(\frac{\frac{8+3}{6}}{\frac{2}{6}}\right)$$

$$= \tan^{-1}\left(\frac{11}{6} \times \frac{6}{2}\right)$$

$$= \tan^{-1}\left(\frac{11}{2}\right)$$

$$= \cot^{-1}\left(\frac{2}{11}\right)$$

Proved.

5. Prove that $2\left(\tan^{-1}1+\tan^{-1}\frac{1}{2}+\tan^{-1}\frac{1}{3}\right)=\pi$ (ISC 2004)

Solution : L.H.S. $=2\left(\tan^{-1}1+\tan^{-1}\frac{1}{2}+\tan^{-1}\frac{1}{3}\right)$

$$=2\left(\frac{\pi}{4}+\tan^{-1}\frac{\frac{1}{2}+\frac{1}{3}}{1-\frac{1}{2}\times\frac{1}{3}}\right)$$

$$=2\left(\frac{\pi}{4}+\tan^{-1}\frac{\frac{5}{6}}{\frac{5}{6}}\right)$$

$$=2\left(\frac{\pi}{4}+\tan^{-1}1\right)$$

$$=2\left(\frac{\pi}{4}+\frac{\pi}{4}\right)$$

$$=2\left(\frac{2\pi}{4}\right)$$

$$=\pi$$

Proved.

6. Prove that $\cot\left(\frac{\pi}{4}-2\cot^{-1}3\right)=7$ (ISC 2005)

Solution : **L.H.S.** $=\cot\left(\frac{\pi}{4}-2\cot^{-1}3\right)$

$$=\cot\left(\frac{\pi}{4}-2\tan^{-1}\frac{1}{3}\right)$$

$$=\cot\left(\frac{\pi}{4}-\tan^{-1}\frac{2.\frac{1}{3}}{1-\left(\frac{1}{3}\right)^2}\right)$$

$$=\cot\left(\frac{\pi}{4}-\tan^{-1}\frac{\frac{2}{3}}{\frac{8}{9}}\right)$$

$$=\cot\left\{\frac{\pi}{4}-\tan^{-1}\left(\frac{2}{3}\times\frac{9}{8}\right)\right\}$$

$$=\cot\left\{\frac{\pi}{4}-\tan^{-1}\left(\frac{3}{4}\right)\right\}$$

$$=\tan\left\{\frac{\pi}{2}-\left(\frac{\pi}{4}-\tan^{-1}\frac{3}{4}\right)\right\}$$

$$=\tan\left\{\frac{\pi}{4}+\tan^{-1}\frac{3}{4}\right\}$$

$$=\frac{\tan\frac{\pi}{4}+\tan\left(\tan^{-1}\frac{3}{4}\right)}{1-\tan\frac{\pi}{4}.\tan\left(\tan^{-1}\frac{3}{4}\right)}$$

$$=\frac{1+\frac{3}{4}}{1-\frac{3}{4}}$$

$$=\frac{\frac{7}{4}}{\frac{1}{4}}$$

$$=7$$

Proved.

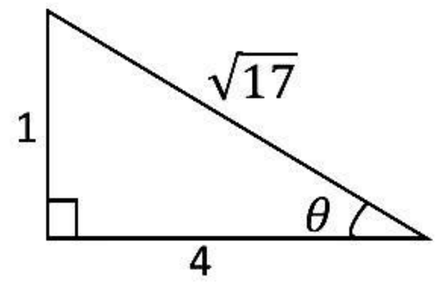
7. Show that $\sin^{-1}\left(\frac{1}{\sqrt{17}}\right) + \cos^{-1}\left(\frac{9}{\sqrt{85}}\right) = \tan^{-1}\left(\frac{1}{2}\right)$ (ISC 2006)

Solution : Let $\sin^{-1}\left(\frac{1}{\sqrt{17}}\right) = \theta$

$\Rightarrow \sin\theta = \frac{1}{\sqrt{17}}$

$\therefore \tan\theta = \frac{1}{4}$

$\therefore \theta = \tan^{-1}\left(\frac{1}{4}\right)$

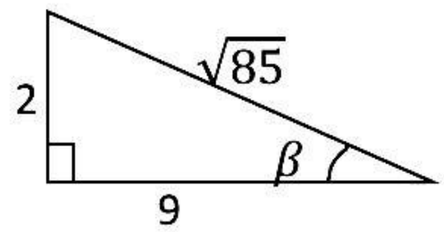


Let $\cos^{-1}\left(\frac{9}{\sqrt{85}}\right) = \beta$

$\therefore \cos\beta = \frac{9}{\sqrt{85}}$

$\therefore \tan\beta = \frac{2}{9}$

$\therefore \beta = \tan^{-1}\left(\frac{2}{9}\right)$



$$\text{L.H.S.} = \sin^{-1}\left(\frac{1}{\sqrt{17}}\right) + \cos^{-1}\left(\frac{9}{\sqrt{85}}\right)$$

$$= \tan^{-1}\left(\frac{1}{4}\right) + \tan^{-1}\left(\frac{2}{9}\right)$$

$$= \tan^{-1}\left(\frac{\frac{1}{4}+\frac{2}{9}}{1-\frac{1}{4}\times\frac{2}{9}}\right)$$

$$= \tan^{-1}\left(\frac{\frac{9+8}{36}}{\frac{36-2}{36}}\right)$$

$$= \tan^{-1}\left(\frac{17}{34}\right)$$

$$= \tan^{-1}\left(\frac{1}{2}\right)$$

Proved.

8. Prove that $2\tan^{-1}\left(\frac{1}{3}\right) + \cot^{-1}4 = \tan^{-1}\left(\frac{16}{13}\right)$ (ISC 2007)

Solution : L.H.S.

$$= 2\tan^{-1}\left(\frac{1}{3}\right) + \cot^{-1}4$$

$$= \tan^{-1}\left(\frac{2.\frac{1}{3}}{1-\left(\frac{1}{3}\right)^2}\right) + \tan^{-1}\frac{1}{4}$$

$$= \tan^{-1}\left(\frac{\frac{2}{3}}{\frac{8}{9}}\right) + \tan^{-1}\frac{1}{4}$$

$$= \tan^{-1}\left(\frac{2}{3}\times\frac{9}{8}\right) + \tan^{-1}\frac{1}{4}$$

$$= \tan^{-1}\frac{3}{4} + \tan^{-1}\frac{1}{4}$$

$$= \tan^{-1}\left(\frac{\frac{3}{4}+\frac{1}{4}}{1-\frac{3}{4}\times\frac{1}{4}}\right)$$

$$= \tan^{-1}\left(\frac{\frac{4}{4}}{\frac{13}{16}}\right)$$

$= \tan^{-1}\left(\frac{16}{13}\right)$ Proved

9. Solve : $\tan^{-1}(x-1) + \tan^{-1} x + \tan^{-1}(x+1) = \tan^{-1} 3x$ (ISC 2008)

Solution : $\tan^{-1}(x-1) + \tan^{-1}(x+1) = \tan^{-1} 3x - \tan^{-1} x$

$\Rightarrow \quad \tan^{-1}\left(\frac{x-1+x+1}{1-(x-1)(x+1)}\right) = \tan^{-1}\left(\frac{3x-x}{1+3x.x}\right)$

$\Rightarrow \quad \frac{2x}{1-(x^2-1)} = \frac{2x}{1+3x^2}$

$\Rightarrow \quad \frac{2x}{2-x^2} = \frac{2x}{1+3x^2}$

$\Rightarrow \quad 2x(1+3x^2) = 2x(2-x^2)$

$\Rightarrow \quad 2x(1+3x^2) - 2x(2-x^2) = 0$

$\Rightarrow \quad 2x(1+3x^2-2+x^2) = 0$

$\Rightarrow \quad x(4x^2-1) = 0$

$\Rightarrow \quad x(2x+1)(2x-1) = 0$

$\Rightarrow \quad x = 0, +\frac{1}{2}, -\frac{1}{2}$ Answer

10. If $\sin^{-1} x + \sin^{-1} y + \sin^{-1} z = \pi$, prove that $x^2 - y^2 - z^2 + 2yz\sqrt{1-x^2} = 0$ (ISC 2009)

Solution : $\sin^{-1} x + \sin^{-1} y + \sin^{-1} z = \pi$

$\Rightarrow \quad \sin^{-1} x + \sin^{-1} y = \pi - \sin^{-1} z$

$\Rightarrow \quad \sin^{-1}\left(x\sqrt{1-y^2} + y\sqrt{1-x^2}\right) = \pi - \sin^{-1} z$

$\Rightarrow \quad x\sqrt{1-y^2} + y\sqrt{1-x^2} = \sin(\pi - \sin^{-1} z)$

$\Rightarrow \quad x\sqrt{1-y^2} + y\sqrt{1-x^2} = \sin(\sin^{-1} z)$

$\Rightarrow \quad x\sqrt{1-y^2} = z - y\sqrt{1-x^2}$

$\Rightarrow \quad \left(x\sqrt{1-y^2}\right)^2 = \left(z - y\sqrt{1-x^2}\right)^2$

$\Rightarrow \quad x^2(1-y^2) = z^2 + y^2(1-x^2) - 2yz\sqrt{1-x^2}$

$\Rightarrow \quad x^2 - x^2y^2 = z^2 + y^2 - y^2x^2 - 2yz\sqrt{1-x^2}$

$\Rightarrow \quad x^2 - y^2 - z^2 + 2yz\sqrt{1-x^2} = 0$ Proved.

11. Solve : $\cos^{-1}[\sin(\cos^{-1} x)] = \frac{\pi}{3}$ (ISC 2010)

Solution : $\cos^{-1}[\sin(\cos^{-1} x)] = \frac{\pi}{3}$

$$\Rightarrow \quad \sin(\cos^{-1} x) = \cos\left(\frac{\pi}{3}\right)$$

$$\Rightarrow \quad \sin\left(\sin^{-1}\sqrt{1-x^2}\right) = \frac{1}{2}$$

$$\Rightarrow \quad \sqrt{1-x^2} = \frac{1}{2}$$

$$\Rightarrow \quad 1 - x^2 = \frac{1}{4}$$

$$\Rightarrow \quad x^2 = 1 - \frac{1}{4}$$

$$\Rightarrow \quad x^2 = \frac{3}{4}$$

$$\Rightarrow \quad x = \pm\frac{\sqrt{3}}{2}$$

Answer.

12. Prove that $\sin\left[2\tan^{-1}\left(\frac{3}{5}\right) - \sin^{-1}\left(\frac{7}{25}\right)\right] = \frac{304}{425}$ (ISC 2010)

Solution : L. H. S.

$$= \sin\left[2\tan^{-1}\left(\frac{3}{5}\right) - \sin^{-1}\left(\frac{7}{25}\right)\right]$$

$$= \sin\left[\tan^{-1}\left(\frac{2.\frac{3}{5}}{1-\left(\frac{3}{5}\right)^2}\right) - \sin^{-1}\left(\frac{7}{25}\right)\right]$$

$$= \sin\left[\tan^{-1}\left(\frac{\frac{6}{5}}{\frac{16}{25}}\right) - \sin^{-1}\left(\frac{7}{25}\right)\right]$$

$$= \sin\left[\tan^{-1}\left(\frac{6}{5}\times\frac{25}{16}\right) - \sin^{-1}\left(\frac{7}{25}\right)\right]$$

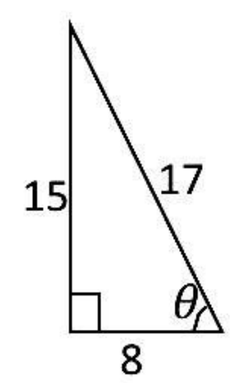
$$= \sin\left[\tan^{-1}\left(\frac{15}{8}\right) - \sin^{-1}\left(\frac{7}{25}\right)\right]$$

Let $\tan^{-1}\left(\frac{15}{8}\right) = \theta$

$$\Rightarrow \quad \tan\theta = \frac{15}{8}$$

$$\Rightarrow \quad \sin\theta = \frac{15}{17}$$

$$\Rightarrow \quad \theta = \sin^{-1}\left(\frac{15}{17}\right)$$



$$\text{L.H.S.} = \sin\left[\sin^{-1}\left(\frac{15}{17}\right) - \sin^{-1}\left(\frac{7}{25}\right)\right]$$

$$= \sin\left[\sin^{-1}\left(\frac{15}{17}\sqrt{1-\left(\frac{7}{25}\right)^2}\right) - \left(\frac{7}{25}\sqrt{1-\left(\frac{15}{17}\right)^2}\right)\right]$$

$$= \frac{15}{17}\sqrt{1-\frac{49}{625}} - \frac{7}{25}\sqrt{1-\frac{225}{289}}$$

$$= \frac{15}{17}\times\sqrt{\frac{576}{625}} - \frac{7}{25}\times\sqrt{\frac{64}{289}}$$

$$= \frac{15}{17}\times\frac{24}{25} - \frac{7}{25}\times\frac{8}{17}$$

$$= \frac{360-56}{17\times 25}$$

$$= \frac{304}{425}$$

Proved.

13. Solve for x : $\sin(2\tan^{-1}x) = 1$ (ISC 2011)

Solution : $\sin(2\tan^{-1}x) = 1$

$$\Rightarrow \quad \sin(2\tan^{-1}x) = \sin\frac{\pi}{2}$$

$$\Rightarrow \quad 2\tan^{-1}x = \frac{\pi}{2}$$

$$\Rightarrow \quad \tan^{-1}x = \frac{\pi}{4}$$

$$\Rightarrow \quad x = \tan\frac{\pi}{4}$$

$$\Rightarrow \quad x = 1$$

Answer.

14. Prove that $2\tan^{-1}\left(\frac{1}{5}\right) + \cos^{-1}\left(\frac{7}{5\sqrt{2}}\right) + 2\tan^{-1}\left(\frac{1}{8}\right) = \frac{\pi}{4}$ (ISC 2011)

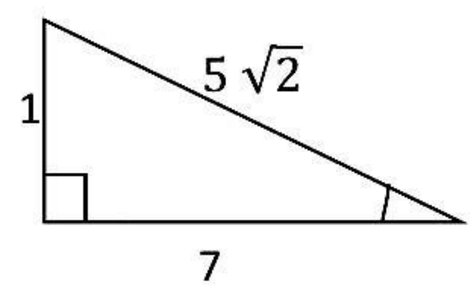
Solution : <u>L.H.S.</u> $= 2\tan^{-1}\left(\frac{1}{5}\right) + \cos^{-1}\left(\frac{7}{5\sqrt{2}}\right) + 2\tan^{-1}\left(\frac{1}{8}\right)$

$$= 2\left\{\tan^{-1}\left(\frac{1}{5}\right) + \tan^{-1}\left(\frac{1}{8}\right)\right\} + \cos^{-1}\left(\frac{7}{5\sqrt{2}}\right)$$

$$= 2\tan^{-1}\left(\frac{\frac{1}{5}+\frac{1}{8}}{1-\frac{1}{5}\cdot\frac{1}{8}}\right) + \cos^{-1}\left(\frac{7}{5\sqrt{2}}\right)$$

$$= 2\tan^{-1}\left(\frac{\frac{13}{40}}{\frac{39}{40}}\right) + \cos^{-1}\left(\frac{7}{5\sqrt{2}}\right)$$

$$= 2\tan^{-1}\left(\frac{13}{39}\right) + \cos^{-1}\left(\frac{7}{5\sqrt{2}}\right)$$



$$= 2\tan^{-1}\left(\frac{1}{3}\right) + \tan^{-1}\left(\frac{1}{7}\right)$$

$$= \tan^{-1}\left(\frac{2.\frac{1}{3}}{1-\left(\frac{1}{3}\right)^2}\right) + \tan^{-1}\left(\frac{1}{7}\right)$$

$$= \tan^{-1}\left(\frac{\frac{2}{3}}{\frac{8}{9}}\right) + \tan^{-1}\left(\frac{1}{7}\right)$$

$$= \tan^{-1}\left(\frac{2}{3}\times\frac{9}{8}\right) + \tan^{-1}\left(\frac{1}{7}\right)$$

$$= \tan^{-1}\left(\frac{3}{4}\right) + \tan^{-1}\left(\frac{1}{7}\right)$$

$$= \tan^{-1}\left(\frac{\frac{3}{4}+\frac{1}{7}}{1-\frac{3}{4}\cdot\frac{1}{7}}\right)$$

$$= \tan^{-1}\left(\frac{\frac{21+4}{28}}{\frac{25}{28}}\right)$$

$$= \tan^{-1}1$$

$$= \frac{\pi}{4}$$

Proved.

15. Prove that : $sec^2(\tan^{-1}2) + cosec^2(\cot^{-1}3) = 15$ (ISC 2012)

Solution : Let $\tan^{-1}2 = \alpha$ and $\cot^{-1}3 = \beta$

$\therefore \quad \tan\alpha = 2$ and $\cot\beta = 3$

L.H.S. $= sec^2\alpha + cosec^2\beta$

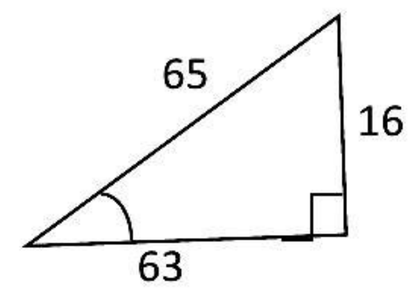
$$= 1 + tan^2\alpha + 1 + cot^2\beta$$
$$= 1 + 4 + 1 + 9$$
$$= 15$$ Proved

16. Prove that $\cos^{-1}\left(\frac{63}{65}\right) + 2\tan^{-1}\left(\frac{1}{5}\right) = \sin^{-1}\left(\frac{3}{5}\right)$ (ISC 2012)

Solution : L.H.S. $= \cos^{-1}\left(\frac{63}{65}\right) + 2\tan^{-1}\left(\frac{1}{5}\right)$

$$= \cos^{-1}\left(\frac{63}{65}\right) + \tan^{-1}\left(\frac{2.\frac{1}{5}}{1-\left(\frac{1}{5}\right)^2}\right)$$

$$= \tan^{-1}\left(\frac{16}{63}\right) + \tan^{-1}\left(\frac{2}{5} \times \frac{25}{24}\right)$$

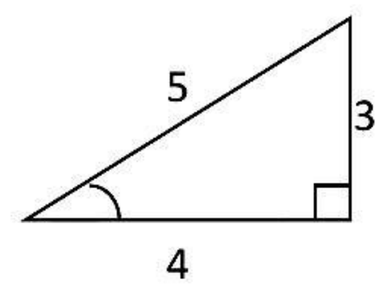


$$= \tan^{-1}\left(\frac{16}{63}\right) + \tan^{-1}\left(\frac{5}{12}\right)$$

$$= \tan^{-1}\left(\frac{\frac{16}{63}+\frac{5}{12}}{1-\frac{16}{63}\times\frac{5}{12}}\right)$$

$$= \tan^{-1}\left(\frac{\frac{192+315}{63\times12}}{\frac{756-80}{63\times12}}\right)$$

$$= \tan^{-1}\left(\frac{507}{676}\right)$$



$$= \tan^{-1}\left(\frac{3}{4}\right)$$

$$= \sin^{-1}\left(\frac{3}{5}\right)$$ Proved.

17. Prove that : $\tan^{-1}\left(\frac{1}{4}\right) + \tan^{-1}\left(\frac{2}{9}\right) = \frac{1}{2}\sin^{-1}\left(\frac{4}{5}\right)$ (ISC 2013)

Solution : L.H.S. $= \tan^{-1}\left(\frac{1}{4}\right) + \tan^{-1}\left(\frac{2}{9}\right)$

$$= \tan^{-1}\left(\frac{\frac{1}{4}+\frac{2}{9}}{1-\frac{1}{4}\times\frac{2}{9}}\right)$$

$$= \tan^{-1}\left(\frac{\frac{9+8}{36}}{\frac{36-2}{36}}\right)$$

$$= \tan^{-1}\left(\frac{17}{34}\right)$$

$$= \tan^{-1}\left(\frac{1}{2}\right)$$

$$= \frac{1}{2}.2\tan^{-1}\left(\frac{1}{2}\right)$$

$$= \frac{1}{2}\sin^{-1}\left(\frac{2.\frac{1}{2}}{1+\left(\frac{1}{2}\right)^2}\right)$$

$$= \frac{1}{2}\sin^{-1}\left(\frac{1}{\frac{5}{4}}\right)$$

$$= \frac{1}{2}\sin^{-1}\left(\frac{4}{5}\right)$$ Proved .

18. Solve for x : $\sin^{-1}x + \sin^{-1}(1-x) = \cos^{-1}x$ (ISC 2013, 2017)

Solution : $\sin^{-1}x + \sin^{-1}(1-x) = \cos^{-1}x$

$\Rightarrow \quad \sin^{-1}x + \sin^{-1}(1-x) = \frac{\pi}{2} - \sin^{-1}x$

$\Rightarrow \quad \sin^{-1}(1-x) = \frac{\pi}{2} - 2\sin^{-1}x$

$\Rightarrow \quad (1-x) = \sin\left(\frac{\pi}{2} - 2\sin^{-1}x\right)$

$\Rightarrow \quad 1-x = \cos(2\sin^{-1}x)$ Let $\sin^{-1}x = \theta$

$\Rightarrow \quad 1-x = \cos 2\theta$ $\therefore \quad \sin\theta = x$

$\Rightarrow \quad 1-x = 1-2\sin^2\theta$

$\Rightarrow \quad 1-x = 1-2x^2$

$\Rightarrow \quad x = 2x^2$

$\Rightarrow \quad 2x^2 - x = 0$

$\Rightarrow \quad x(2x-1) = 0$

$\Rightarrow \quad x = 0 \text{ or } \frac{1}{2}$ Answer.

19. Evaluate : $\tan\left[2\tan^{-1}\frac{1}{2} - \cot^{-1}3\right]$ (ISC 2014)

Solution : $\tan\left[2\tan^{-1}\frac{1}{2} - \cot^{-1}3\right]$

$= \tan\left[\tan^{-1}\left\{\frac{2.\frac{1}{2}}{1-\left(\frac{1}{2}\right)^2}\right\} - \tan^{-1}\left(\frac{1}{3}\right)\right]$

$= \tan\left[\tan^{-1}\left(\frac{1}{\frac{3}{4}}\right) - \tan^{-1}\left(\frac{1}{3}\right)\right]$

$= \tan\left[\tan^{-1}\left(\frac{4}{3}\right) - \tan^{-1}\left(\frac{1}{3}\right)\right]$

$= \tan\left[\tan^{-1}\left(\frac{\frac{4}{3}-\frac{1}{3}}{1+\frac{4}{3}\times\frac{1}{3}}\right)\right]$

$= \frac{\frac{3}{3}}{\frac{13}{9}}$

$= \frac{9}{13}$ Answer

20. If $\cos^{-1}x + \cos^{-1}y + \cos^{-1}z = \pi$, prove that $x^2 + y^2 + z^2 + 2xyz = 1$. (ISC 2014)

Solution : $\cos^{-1}x + \cos^{-1}y + \cos^{-1}z = \pi$

$\Rightarrow \quad \cos^{-1}x + \cos^{-1}y = \pi - \cos^{-1}z$

$\Rightarrow \quad \cos^{-1}\left(xy - \sqrt{1-x^2}\sqrt{1-y^2}\right) = \pi - \cos^{-1}z$

$\Rightarrow \quad xy - \sqrt{1-x^2}\sqrt{1-y^2} = \cos(\pi - \cos^{-1}z)$

$\Rightarrow \quad xy - \sqrt{1-x^2}\sqrt{1-y^2} = -\cos(\cos^{-1}z)$

$\Rightarrow \quad xy - \sqrt{1-x^2}\sqrt{1-y^2} = -z$

$\Rightarrow \quad xy + z = \sqrt{1-x^2}\sqrt{1-y^2}$

$$\Rightarrow \quad (xy+z)^2 = \left(\sqrt{1-x^2}\sqrt{1-y^2}\right)^2$$
$$\Rightarrow \quad x^2y^2 + z^2 + 2xyz = (1-x^2)(1-y^2)$$
$$\Rightarrow \quad x^2y^2 + z^2 + 2xyz = 1 - x^2 - y^2 + x^2y^2$$
$$\Rightarrow \quad x^2 + y^2 + z^2 + 2xyz = 1$$

Proved.

21. Solve : $\cos^{-1}\{\sin(\cos^{-1}x)\} = \frac{\pi}{6}$ (ISC 2015)

Solution : $\cos^{-1}\{\sin(\cos^{-1}x)\} = \frac{\pi}{6}$

$$\Rightarrow \quad \sin(\cos^{-1}x) = \cos\left(\frac{\pi}{6}\right)$$
$$\Rightarrow \quad \sin\left(\sin^{-1}\sqrt{1-x^2}\right) = \frac{\sqrt{3}}{2}$$
$$\Rightarrow \quad \sqrt{1-x^2} = \frac{\sqrt{3}}{2}$$
$$\Rightarrow \quad 1 - x^2 = \frac{3}{4}$$
$$\Rightarrow \quad 4 - 4x^2 = 3$$
$$\Rightarrow \quad 4x^2 = 1$$
$$\Rightarrow \quad x^2 = \frac{1}{4}$$
$$\Rightarrow \quad x = \pm\frac{1}{2}$$

Answer

22. Solve for x : $\sin^{-1}\left(\frac{5}{x}\right) + \sin^{-1}\left(\frac{12}{x}\right) = \frac{\pi}{2}$, $x \neq 0$. (ISC 2015)

Solution : $\sin^{-1}\left(\frac{5}{x}\right) + \sin^{-1}\left(\frac{12}{x}\right) = \frac{\pi}{2}$

$$\Rightarrow \quad \sin^{-1}\left(\frac{12}{x}\right) = \frac{\pi}{2} - \sin^{-1}\left(\frac{5}{x}\right)$$
$$\Rightarrow \quad \frac{12}{x} = \sin\left\{\frac{\pi}{2} - \sin^{-1}\left(\frac{5}{x}\right)\right\}$$
$$\Rightarrow \quad \frac{12}{x} = \cos\left\{\sin^{-1}\left(\frac{5}{x}\right)\right\}$$
$$\Rightarrow \quad \frac{12}{x} = \cos\left\{\cos^{-1}\sqrt{1-\left(\frac{5}{x}\right)^2}\right\}$$
$$\Rightarrow \quad \frac{12}{x} = \sqrt{1-\frac{25}{x^2}}$$
$$\Rightarrow \quad \left(\frac{12}{x}\right)^2 = \left(\sqrt{1-\frac{25}{x^2}}\right)^2$$
$$\Rightarrow \quad \frac{144}{x^2} = 1 - \frac{25}{x^2}$$
$$\Rightarrow \quad \frac{144}{x^2} + \frac{25}{x^2} = 1$$
$$\Rightarrow \quad \frac{169}{x^2} = 1$$
$$\Rightarrow \quad x^2 = 169$$
$$\Rightarrow \quad x = \pm 13$$

Answer $x = 13$

23. Solve for x : $\tan(\cos^{-1} x) = \frac{2}{\sqrt{5}}$ (ISC 2016)

Solution : $\tan(\cos^{-1} x) = \frac{2}{\sqrt{5}}$

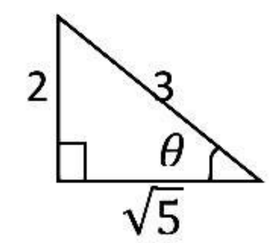
$\Rightarrow \quad \cos^{-1} x = \tan^{-1}\left(\frac{2}{\sqrt{5}}\right)$

$\Rightarrow \quad \cos^{-1} x = \cos^{-1}\left(\frac{\sqrt{5}}{3}\right)$

$\Rightarrow \quad x = \frac{\sqrt{5}}{3}$ Answer.

Let $\tan^{-1}\left(\frac{2}{\sqrt{5}}\right) = \theta$

$\therefore \quad \tan\theta = \frac{2}{\sqrt{5}}$



24. If $\sin^{-1} x + \tan^{-1} x = \frac{\pi}{2}$, prove that $2x^2 + 1 = \sqrt{5}$ (ISC 2016)

Solution : $\sin^{-1} x + \tan^{-1} x = \frac{\pi}{2}$

$\Rightarrow \quad \tan^{-1} x = \frac{\pi}{2} - \sin^{-1} x$

$\Rightarrow \quad \tan^{-1} x = \cos^{-1} x$

$\Rightarrow \quad \tan^{-1} x = \tan^{-1}\left(\frac{\sqrt{1-x^2}}{x}\right)$

$\Rightarrow \quad x = \frac{\sqrt{1-x^2}}{x}$

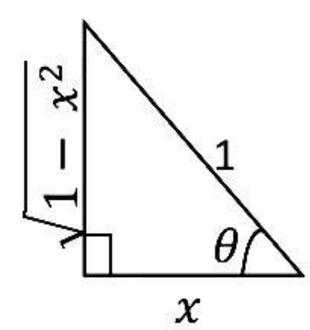
$\Rightarrow \quad x^2 = \sqrt{1-x^2}$

$\Rightarrow \quad x^4 = 1 - x^2$

$\Rightarrow \quad (x^2)^2 + x^2 - 1 = 0$

Let $\cos^{-1} x = \theta$

$\therefore \cos\theta = x$



$\Rightarrow \quad x^2 = \frac{-1 \pm \sqrt{1-4(-1)}}{2}$

$\Rightarrow \quad x^2 = \frac{-1 \pm \sqrt{5}}{2}$

$\Rightarrow \quad x^2 = \frac{-1 + \sqrt{5}}{2}$ (As x^2 is positive)

$\Rightarrow \quad 2x^2 = -1 + \sqrt{5}$

$\Rightarrow \quad 2x^2 + 1 = \sqrt{5}$ Proved

25. Prove that $\frac{1}{2}\cos^{-1}\left(\frac{1-x}{1+x}\right) = \tan^{-1}\sqrt{x}$ (ISC 2017)

Solution : R.H.S. $= \tan^{-1}\sqrt{x}$

$= \frac{1}{2}.2\tan^{-1}\sqrt{x}$

$= \frac{1}{2}\cos^{-1}\left(\frac{1-(\sqrt{x})^2}{1+(\sqrt{x})^2}\right)$

$= \frac{1}{2}\cos^{-1}\left(\frac{1-x}{1+x}\right)$ Proved

26. Solve for x : $\sin^{-1} x + \sin^{-1}(1-x) = \cos^{-1} x$, $x \neq 0$ (ISC 2017)

Solution : $\sin^{-1} x + \sin^{-1}(1-x) = \cos^{-1} x$

$\Rightarrow \quad \sin^{-1} x + \sin^{-1}(1-x) = \frac{\pi}{2} - \sin^{-1} x$

$\Rightarrow \quad \sin^{-1}(1-x) = \frac{\pi}{2} - 2\sin^{-1}x$

$\Rightarrow \quad 1 - x = \sin\left(\frac{\pi}{2} - 2\sin^{-1}x\right)$

$\Rightarrow \quad 1 - x = \cos(2\sin^{-1}x)$ Let $\sin^{-1}x = \theta$

$\Rightarrow \quad 1 - x = \cos 2\theta$ $\therefore \sin\theta = x$

$\Rightarrow \quad 1 - x = 1 - 2\sin^2\theta$

$\Rightarrow \quad 1 - x = 1 - 2x^2$

$\Rightarrow \quad x = 2x^2$

$\Rightarrow \quad 2x^2 - x = 0$

$\Rightarrow \quad x(2x-1) = 0$

$\Rightarrow \quad x = 0, \frac{1}{2}$ Answer

27. Solve for x : $3\tan^{-1}x + \cot^{-1}x = \pi$ (ISC 2018)

Solution : $3\tan^{-1}x + \cot^{-1}x = \pi$

$\Rightarrow \quad 3\tan^{-1}x + \frac{\pi}{2} - \tan^{-1}x = \pi$

$\Rightarrow \quad 2\tan^{-1}x = \frac{\pi}{2}$

$\Rightarrow \quad \tan^{-1}x = \frac{\pi}{4}$

$\Rightarrow \quad x = \tan\left(\frac{\pi}{4}\right)$

$\Rightarrow \quad x = 1$ Answer

28. If $\tan^{-1}a + \tan^{-1}b + \tan^{-1}c = \pi$, prove that $a + b + c = abc$ (ISC 2018)

Solution : $\tan^{-1}a + \tan^{-1}b + \tan^{-1}c = \pi$

$\Rightarrow \quad \tan^{-1}a + \tan^{-1}b = \pi - \tan^{-1}c$

$\Rightarrow \quad \tan^{-1}\left(\frac{a+b}{1-ab}\right) = \pi - \tan^{-1}c$

$\Rightarrow \quad \frac{a+b}{1-ab} = \tan(\pi - \tan^{-1}c)$

$\Rightarrow \quad \frac{a+b}{1-ab} = -\tan(\tan^{-1}c)$

$\Rightarrow \quad \frac{a+b}{1-ab} = -c$

$\Rightarrow \quad a + b = -c + abc$

$\Rightarrow \quad a + b + c = abc$ Proved

29. Solve for x : $\tan^{-1}\left(\frac{x-1}{x-2}\right) + \tan^{-1}\left(\frac{x+1}{x+2}\right) = \frac{\pi}{4}$ (ISC 2019)

Solution : $\tan^{-1}\left(\frac{x-1}{x-2}\right) + \tan^{-1}\left(\frac{x+1}{x+2}\right) = \frac{\pi}{4}$

$\Rightarrow \quad \tan^{-1}\left\{\frac{\frac{x-1}{x-2} + \frac{x+1}{x+2}}{1 - \frac{x-1}{x-2} \times \frac{x+1}{x+2}}\right\} = \frac{\pi}{4}$

$\Rightarrow \quad \tan^{-1}\left\{\frac{\frac{(x-1)(x+2) + (x-2)(x+1)}{(x-2)(x+2)}}{\frac{(x-2)(x+2) - (x-1)(x+1)}{(x-2)(x+2)}}\right\} = \frac{\pi}{4}$

$$\Rightarrow \quad \frac{x^2+x-2+x^2-x-2}{x^2-4-x^2+1} = \tan\left(\frac{\pi}{4}\right)$$

$$\Rightarrow \quad \frac{2x^2-4}{-3} = 1$$

$$\Rightarrow \quad 2x^2 - 4 = -3$$

$$\Rightarrow \quad 2x^2 = 1$$

$$\Rightarrow \quad x^2 = \frac{1}{2}$$

$$\Rightarrow \quad x = \pm\frac{1}{\sqrt{2}}$$ Answer

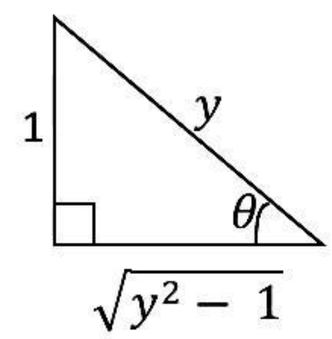
30. If $\sec^{-1}x = \operatorname{cosec}^{-1}y$, show that $\frac{1}{x^2} + \frac{1}{y^2} = 1$ (ISC 2019)

Solution : Let $\operatorname{cosec}^{-1}y = \theta$

$\therefore \operatorname{cosec}\theta = y$

$\therefore \sec\theta = \frac{y}{\sqrt{y^2-1}}$

$\therefore \theta = \sec^{-1}\left(\frac{y}{\sqrt{y^2-1}}\right)$



Given $\sec^{-1}x = \operatorname{cosec}^{-1}y$

$$\Rightarrow \quad \sec^{-1}x = \sec^{-1}\left(\frac{y}{\sqrt{y^2-1}}\right)$$

$$\Rightarrow \quad x = \frac{y}{\sqrt{y^2-1}}$$

$$\Rightarrow \quad x^2 = \frac{y^2}{y^2-1}$$

$$\Rightarrow \quad \frac{1}{x^2} = \frac{y^2-1}{y^2}$$

$$\Rightarrow \quad \frac{1}{x^2} = 1 - \frac{1}{y^2}$$

$$\Rightarrow \quad \frac{1}{x^2} + \frac{1}{y^2} = 1$$ Proved

31. Prove that : $tan^2(\sec^{-1}2) + cot^2(\operatorname{cosec}^{-1}3) = 11$ (ISC 2020)

Solution : Let $\sec^{-1}2 = \theta \Rightarrow \sec\theta = 2$

And $\operatorname{cosec}^{-1}3 = \beta \Rightarrow \operatorname{cosec}\beta = 3$

$\therefore$ L.H.S. $= tan^2(\sec^{-1}2) + cot^2(\operatorname{cosec}^{-1}\beta)$

$= tan^2\theta + cot^2\beta$

$= sec^2\theta - 1 + cosec^2\beta - 1$

$= 2^2 - 1 + 3^2 - 1$

$= 11$ Proved

32. If $\cos^{-1}\frac{x}{2} + \cos^{-1}\frac{y}{3} = \theta$, then prove that (ISC 2020)

$$9x^2 - 12xy\cos\theta + 4y^2 = 36\sin^2\theta$$

Solution : Given $\cos^{-1}\frac{x}{2} + \cos^{-1}\frac{y}{3} = \theta$

$$\Rightarrow \quad \cos^{-1}\left(\frac{x}{2}\cdot\frac{y}{3} - \sqrt{1-\left(\frac{x}{2}\right)^2}\sqrt{1-\left(\frac{y}{3}\right)^2}\right) = \theta$$

$$\Rightarrow \quad \frac{xy}{6} - \sqrt{1-\frac{x^2}{4}}\sqrt{1-\frac{y^2}{9}} = \cos\theta$$

$$\Rightarrow \quad \frac{xy}{6} - \cos\theta = \sqrt{1-\frac{x^2}{4}}\sqrt{1-\frac{y^2}{9}}$$

$$\Rightarrow \quad \left(\frac{xy}{6} - \cos\theta\right)^2 = \left(1-\frac{x^2}{4}\right)\left(1-\frac{y^2}{9}\right)$$

$$\Rightarrow \quad \frac{x^2y^2}{36} + cos^2\theta - \frac{2\,xy}{6}\cos\theta = 1 - \frac{x^2}{4} - \frac{y^2}{9} + \frac{x^2y^2}{36}$$

Multiplying each term by 36

$$\Rightarrow \quad x^2y^2 + 36\,cos^2\theta - 12xy\,\cos\theta = 36 - 9\,x^2 - 4y^2 + x^2y^2$$

$$\Rightarrow \quad 9x^2 - 12xy\,\cos\theta + 4y^2 = 36 - 36\,cos^2\theta$$

$$\Rightarrow \quad 9x^2 - 12xy\,\cos\theta + 4y^2 = 36(1 - cos^2\theta)$$

$$\Rightarrow \quad 9\,x^2 - 12\,xy\,\cos\theta + 4y^2 = 36\,sin^2\theta$$ Proved

33. Evaluate : $\cos(2\cos^{-1}x + \sin^{-1}x)$ at $x = \frac{1}{5}$. (ISC 2020)

Solution : $\cos(2\cos^{-1}x + \sin^{-1}x)$

$$= \cos\{\cos^{-1}x + (\cos^{-1}x + \sin^{-1}x)\}$$

$$= \cos\left(\cos^{-1}x + \frac{\pi}{2}\right)$$

$$= \cos\left(\frac{\pi}{2} + \cos^{-1}x\right)$$

$$= -\sin(\cos^{-1}x)$$

$$= -\sin\left(\sin^{-1}\sqrt{1-x^2}\right)$$

$$= -\sqrt{1-x^2}$$

$$= -\sqrt{1-\frac{1}{25}} \qquad \text{at} \quad x = \frac{1}{5}$$

$$= -\sqrt{\frac{24}{25}}$$

$$= -\frac{2\sqrt{6}}{5}$$ Answer